

MATHEMATICS

Time Allowed: 70 minutes.

Calculators may be used.

Set your work out clearly. Use a new page for each question.

Question 1. (12 marks)

- (a) Factorize $27a^3 - 125b^6$
- (b) Expand $4(r - 6)^2$
- (c) Write the derivatives of:
 - (i) $\frac{1}{4x^4}$
 - (ii) $\frac{1}{2x + 1}$
 - (iii) $\sqrt{x^3}$
 - (iv) $(3x^2 - 5x - 2)(1 + x^3)^{-3}$

Question 2. (12 marks)

- (a) Write down the functions whose derivatives are, respectively,
 - (i) $\frac{1}{x^2}$
 - (ii) $3x^2 + 2$
 - (iii) x^4
- (b) Find the equation of the line through the point (2, 0) and the intersection of the two lines : $x + y = 0$
 $2x - y = 3$.
- (c) Find the point of intersection of the parabola $x^2 = 4y$ with the line $y = 2x - 4$.

Question 3. (12 marks)

For the function $y = x^3 - 3x^2 - 9x + 30$.

- (i) Find the stationary points of $y = x^3 - 3x^2 - 9x + 30$. Establish the nature of these points with the use of calculus
- (ii) Sketch the curve, showing all important features.
- (iii) Find the equations of the tangent and the normal to the curve $y = x^3 - 3x^2 - 9x + 30$ at the point (2, 8).
- (iv) Find the co-ordinates of the points (or point) where the line $y = 30 - 9x$ intersects (or touches) the curve.

Question 4. (12 marks)

- (a) The first four terms of three sequences are given below:
 A: 3, 1, 4, 5; B: 4, -2, 1, -1/2; C: -3, -1, 1, 3.
 (i) In each case state whether the sequence is geometric, arithmetic or neither.
 (ii) For the sequence C, assume that it is, in fact, the type you consider possible and write down :
 1. the tenth term
 and 2. the sum of the first ten terms.
- (b) In the sequence 5, 11, 29, the n th term is $3^n + 2$.
 Write down (i) the fourth term
 (ii) the sum of the first n terms.

Continued over

Question 5. (12 marks)

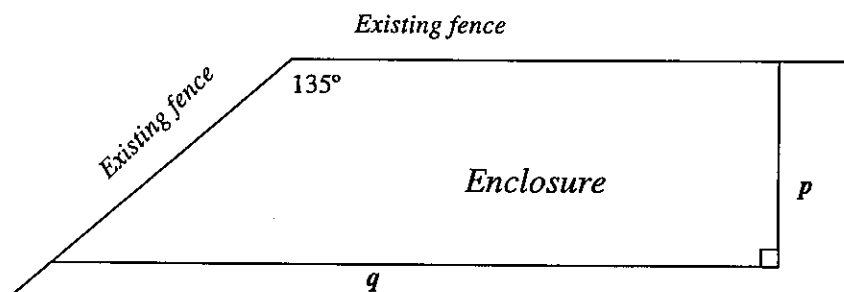
- (a) (i) The series $1/3 - 1/6 + 1/12$ is geometric.

Find the sum of the first ten terms. Give your answer as a **rational** number in its lowest terms.

- (ii) Can there be an infinite geometric series with a limiting sum of $5/8$ and the first term of 2?

All working and reasoning must be shown.

(b)



At the corner of the property an enclosure is to be made for a chicken enclosure. The existing fence, which has an angle of 135° , is used for two sides. 170 metres of fencing is available for use to make the enclosure.

Show that the area is $A = 170p - 3/2 p^2$ and show that the area of the enclosure is a maximum when $q = 2p$.

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$$1(a) 27a^3 - 125b^6$$

$$= (3a)^3 - (5b^2)^3$$

$$(2) = (3a - 5b^2)(9a^2 + 15ab^2 + 25b^4)$$

$$(b) 4(r-6)^2$$

$$= 4(r^2 - 12r + 36)$$

$$(2) = 4r^2 - 48r + 144$$

$$(c)(i) \frac{d}{dx} \left(\frac{1}{4} x^{-4} \right)$$

$$= -x^{-5}$$

$$= -\frac{1}{x^5}$$

(2)

$$(ii) \frac{d}{dx} (2x+1)^{-1}$$

$$= -1(2x+1)^{-2} \cdot 2$$

$$= -2$$

$$(2) \frac{d}{dx} (2x+1)^{-2}$$

$$(iii) \frac{d}{dx} (x^{3/2})$$

$$= \frac{3}{2} x^{1/2}$$

$$(2) = \frac{3\sqrt{x}}{2}$$

$$(iv) \frac{d}{dx} (3x^2 - 5x - 2)(1+x^3)$$

$$= (1+x^3)^{-3} \cdot (6x-5) + (3x^2-5x-2) \cdot (1+x^3)^{-4}$$

$$= (1+x^3)^{-3} [6x-5 - \frac{3x^2-5x-2}{1+x^3}]$$

$$= \frac{(6x-5)(1+x^3) - 3x^2 + 5x + 2}{(1+x^3)^4}$$

$$= \frac{6x - 5 + 6x^4 - 5x^3 - 3x^2 + 5x + 2}{(1+x^3)^4}$$

$$= \frac{-2x^5 + 6x^4 + 40x^3 + 18x^2 + 6}{(1+x^3)^4}$$

$$2. (a) (i) \frac{dy}{dx} = x^{-2}$$

$$y = -x^{-1}$$

$$(2) \therefore y = \frac{1}{x} + 1$$

$$(ii) \frac{dy}{dx} = 3x^2 + 2$$

$$(2) y = x^3 + 2x$$

$$(iii) \frac{dy}{dx} = x^4$$

$$(2) y = \frac{1}{5} x^5 +$$

$$2. (b) \begin{aligned} x+y &= 0 \\ 2x-y &= 3 \\ 2x-y-3 &= 0 \\ x+y+k(2x-y-3) &= 0 \end{aligned}$$

$$(20) \begin{aligned} 2+0+k(4-0-3) &= 0 \\ 2+k &= 0 \\ \therefore k &= -2 \end{aligned}$$

$$\begin{aligned} x+y-2(2x-y-3) &= 0 \\ x+y-4x+2y+6 &= 0 \end{aligned}$$

$$(3) \quad -3x+3y+6=0$$

$$\begin{aligned} x-y-2 &= 0 \\ \text{test (2,0)} \quad 2-0-2 &= 0 \end{aligned}$$

$$(c) \begin{aligned} x^2 &= 4y \quad (1) \\ y &= 2x-4 \quad (2) \end{aligned}$$

$$\begin{aligned} (2) \text{ into } (1) \quad x^2 &= 4(2x-4) \\ x^2-8x+16 &= 0 \\ (x-4)^2 &= 0 \\ x &= 4 \end{aligned}$$

$$\text{in } (2) \quad y = 8-4 = 4$$

$$\text{check in } (1) \quad 16 = 4 \cdot 4$$

$$(3) \quad \text{Point of intersection is } (4, 4)$$

it actually touches parabola.

$$2. \quad y = x^3 - 3x^2 - 9x + 30$$

$$\begin{aligned} (i) \quad y' &= 3x^2 - 6x - 9 \\ &= 3(x^2 - 2x - 3) \\ &= 3(x-3)(x+1) \end{aligned}$$

$$\begin{aligned} (1) \quad y' &= 0, \quad x = 3, -1 \\ x=3, y &= 27-27-27+30 \\ &= -3 \end{aligned}$$

$$\begin{aligned} (1) \quad (3, -3) \\ x=-1, y &= -1-3+9+30 \\ &= 35 \end{aligned}$$

$$\begin{aligned} (1) \quad (-1, 35) \\ y'' &= 6x-6 \quad x=-1 \\ y'' &= 6(-1-1) = -12 < 0 \end{aligned}$$

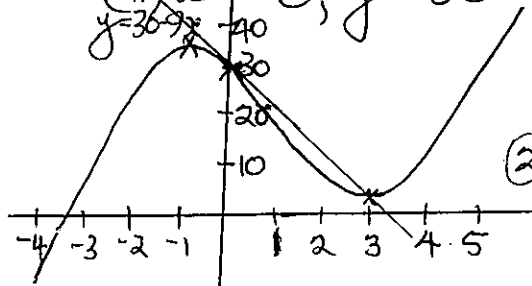
$$\text{at } x=3, y'' = 6(3-1) = 12 > 0$$

$$(1) \therefore \text{minimum at } (3, -3)$$

$$\text{at } x=-1, y'' = 6(-1-1) = -12 < 0$$

$$(1) \therefore \text{maximum at } (-1, 35)$$

$$(ii) \quad x=0, y=30$$



$$f(-2) = -8-12+18+30 > 0$$

$$f(-3) = -27-27+27+30 > 0$$

$$f(-4) = -64-48+36+30 < 0$$

$$(iii) \quad \frac{dy}{dx} = m = 3(x-3)(x+1)$$

$$\begin{aligned} \text{at } x=2, m &= 3(2-3)(2+1) \\ &= -3 \cdot 3 = -9 \end{aligned}$$

$$\therefore m_n = \frac{1}{9}$$

$$\begin{aligned} \text{eqn of tangent: } y-8 &= -9(x-2) \\ y-8 &= -9x+18 \\ 9x+y-26 &= 0 \quad (1) \end{aligned}$$

$$\text{eqn of normal: } y-8 = \frac{1}{9}(x-2)$$

$$\begin{aligned} 9y-72 &= x-2 \\ x-9y+70 &= 0 \quad (1) \end{aligned}$$

$$(iv) \quad y = 30 - 9x = x^3 - 3x^2 - 9x + 30$$

$$0 = x^3 - 3x^2$$

$$x^2(x-3) = 0$$

$$x=0, x=3$$

$$\begin{aligned} \text{line } y &= 30-9x \text{ intersects} \\ \text{with } y &= x^3-3x^2-9x+30 \quad (1) \end{aligned}$$

$$\begin{aligned} \text{at } (3, 3) \text{ and touches (two roots)} \\ \text{it at } (0, 30) \quad (1) \end{aligned}$$

$$4. (a) \quad A \quad 3, 1, 4, 5$$

$$(i) \quad (1) \text{ neither}$$

$$B. \quad 4, -2, 1, -\frac{1}{2}$$

$$\begin{aligned} (1) \quad r &= -\frac{1}{2} \\ \therefore \text{geometric} \\ -3, -1, 1, 3 \\ d &= 2 \end{aligned}$$

$$(1) \text{ arithmetic}$$

$$(ii) \quad \begin{aligned} a &= -3 \\ d &= 2 \end{aligned}$$

$$1. \quad T_n = a + (n-1)d$$

$$\begin{aligned} T_{10} &= a + 9d \\ &= -3 + 18 \\ &= 15 \end{aligned}$$

$$(2) \quad = 15$$

$$2. \quad S_n = \frac{n}{2}(a+l)$$

$$\begin{aligned} &= \frac{10}{2}(-3+15) \\ &= 5 \times 12 \\ &= 60 \end{aligned}$$

$$(2) \quad = 60$$

$$\text{f geometric}$$

$$a = -3$$

$$r = \frac{1}{3}$$

$$1. \quad T_n = ar^{n-1}$$

$$\begin{aligned} T_{10} &= -3\left(\frac{1}{3}\right)^9 \\ &= -\frac{1}{3^8} = -\frac{1}{6561} \end{aligned}$$

$$2. \quad S_n = \frac{a(1-r^n)}{(1-r)}$$

$$4(b) \ 5, 11, 29, \dots 3^n + 2$$

$$T_4 = 3^4 + 2 = 81 + 2 = 83 \quad (2)$$

$$a = 5 \quad r = ? \text{ not arith/geom but both } 3$$

$$3^1 + 2 + 3^2 + 2 + 3^3 + 2 + \dots + 3^n + 2$$

$$= 3^1 + 3^2 + 3^3 + \dots + 3^n + 2 \times n$$

$$S_n \text{ of } 3^1 + 3^2 + \dots \quad (1)$$

$$= \frac{3(3^n - 1)}{3 - 1}$$

$$= \frac{3^{n+1} - 3}{2}$$

$$S_n \text{ of } 2 + 2 + \dots = 2n \quad (1)$$

$$\text{Total sum} = \frac{3^{n+1} - 3}{2} + 2n$$

$$\sum_{n=1}^n 3^n + 2 = \text{or } \frac{3}{2}(3^n - 1) + 2n \quad (1)$$

$$5(a)(1) \quad \frac{1}{3} - \frac{1}{6} + \frac{1}{12}$$

$$r = \frac{-\frac{1}{6}}{\frac{1}{3}} = -\frac{1}{2}$$

$$r = \frac{1}{12} \div -\frac{1}{6} = -\frac{1}{2} \quad (1)$$

$$S_{10} = \frac{\frac{1}{3}(1 - (-\frac{1}{2})^{10})}{1 - (-\frac{1}{2})} \quad (1)$$

$$= \frac{\frac{1}{3}(1 - \frac{1}{1024})}{\frac{3}{2}}$$

$$\frac{1}{3} \times \frac{2}{3} \left(\frac{1023}{1024} \right)$$

$$= \frac{2}{9} \left(\frac{1023}{1024} \right)$$

$$(2) = \frac{1023}{4608} = \frac{341}{1536}$$

$$(ii) \quad \frac{a}{1-r} = \frac{5}{8}$$

$$\frac{2}{1-r} = \frac{5}{8}$$

$$16 = 5 - 5r$$

$$5r = -11$$

$$(2) \quad r = -\frac{11}{5}$$

No, r must be between -1 and +1 i.e.

$$|r| < 1$$

5(b)

$$p + q = 170$$

$$q = 170 - p \quad (1)$$

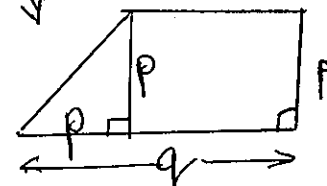
$$\text{Area} = p(q - p) + \frac{1}{2}p^2$$

$$= p(170 - p - p) + \frac{1}{2}p^2$$

$$= p(170 - 2p) + \frac{1}{2}p^2$$

$$= 170p - 2p^2 + \frac{1}{2}p^2$$

$$= 170p - \frac{3}{2}p^2 \quad (1)$$



$$\text{Area} = \triangle + \square$$

$$= \frac{1}{2}p^2 + (q - p)p$$

$$= \frac{1}{2}p^2 + pq - p^2$$

$$A' = 170 - 3p$$

$$A' = 0, \quad 3p = 170$$

$$p = \frac{170}{3}$$

$$A' = -3 < 0 \quad (1)$$

∴ at $p = 56\frac{2}{3}$ a maximum occurs

$$\text{when } p = 56\frac{2}{3}$$

$$q = 170 - 56\frac{2}{3}$$

$$= 113\frac{1}{3} \quad (1)$$

$$\text{i.e. } q = 2p \quad (1)$$